

AG-512B Laser Alignment Report

Report ALN-AG512B-2026-06 | Instrument LAS-512-09 | Final survey 2026-06-22

1. Instrument and setup

Laser instrument certificate CAL-LAS-512-2026-04 was valid through 2027-04-30. Coupling spacer length was measured as 196.0 mm. The 1488 rpm running-speed reference identifies the intended operating condition only and is not a tachometer acceptance result.

record_id	setup parameter	value	units	status
ALN-SET-01	Instrument certificate	CAL-LAS-512-2026-04		ACCEPTED
ALN-SET-02	Spacer length	196.0	mm	ACCEPTED
ALN-SET-03	Running-speed reference	1488	rpm	CONTEXT

2. Local final-alignment limits

parameter	local limit	operator
Horizontal offset	0.060 mm maximum	<=
Vertical offset	0.050 mm maximum	<=
Horizontal angularity	0.040 mm/100 mm maximum	<=
Vertical angularity	0.040 mm/100 mm maximum	<=
Motor-foot soft foot	0.050 mm maximum	<=

3. Pre-correction readings

record_id	parameter	unrounded result	units	local result	record use
ALN-PRE-01	Horizontal offset	+0.1780	mm	FAIL	Traceability only
ALN-PRE-02	Vertical offset	-0.1460	mm	FAIL	Traceability only
ALN-PRE-03	Horizontal angularity	+0.0920	mm/100 mm	FAIL	Traceability only
ALN-PRE-04	Vertical angularity	-0.0840	mm/100 mm	FAIL	Traceability only

4. Final post-correction readings

record_id	parameter	unrounded result	displayed result	units	operator	limit	result
ALN-FIN-01	Horizontal offset	-0.0600	-0.06	mm	<=	0.060	PASS
ALN-FIN-02	Vertical offset	+0.0512	+0.05	mm	<=	0.050	FAIL
ALN-FIN-03	Horizontal angularity	+0.0386	+0.04	mm/100 mm	<=	0.040	PASS
ALN-FIN-04	Vertical angularity	-0.0400	-0.04	mm/100 mm	<=	0.040	PASS

Acceptance is based on the unrounded result. Equality passes for the recorded <= operator.

5. Final motor soft-foot residuals

record_id	motor foot	final residual	units	operator	limit	result
SF-01	Foot A - DE left	0.021	mm	<=	0.050	PASS
SF-02	Foot B - DE right	0.050	mm	<=	0.050	PASS
SF-03	Foot C - NDE left	0.034	mm	<=	0.050	PASS
SF-04	Foot D - NDE right	0.056	mm	<=	0.050	FAIL

6. Preliminary draft summary

record_id	draft value set	draft status	final disposition
ALN-DRAFT-00	-0.059, +0.050, +0.039, -0.040	PASS	SUPERSEDED by ALN-FIN-01 through ALN-FIN-04

7. Sign-off

Mechanical technician: L. Rivas, signed 2026-06-22. Reliability engineer: P. Chen, signed 2026-06-22.